



CHAPTER

Control Cabinet, PLC and HMI

The IP55 control cabinet, PLC I/O and operator touchscreen

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14.1 Overview and function

The control cabinet houses the PLC, I/O modules, VFDs and terminal wiring that supervise the whole unit: a Weidmüller UC20-WL2000-AC controller (CODESYS/u-OS based) coordinates distributed ADAM I/O modules (ADAM-4024/4050/4017+/5000), the process VFDs (chapter 9), and the dosing subsystem (chapter 12), with an optional touchscreen HMI for local operation.



Figure 14.1. Suggested figure: photograph of the open control cabinet with the PLC, I/O modules and terminal strips labeled.

14.2 Key specifications

- Cabinet ingress protection: IP55.
- Controller: Weidmüller UC20-WL2000-AC (CODESYS/u-OS).
- Distributed I/O: Advantech ADAM series. The **ADAM-3600-C2G** Wireless Intelligent RTU (also usable as the main controller) is fully datasheeted in HydroVolta's library: Cortex-A8 AM3352 @600 MHz, 256 MB DDR3L RAM, RT-Linux 3.12; onboard 8AI/8DI/4DO plus a 4-slot I/O expansion bay (compatible modules: ADAM-3613/3617/3618/3624/3651/3656); 1×RS232/485 (DB9) + 2×RS485 (terminal block), 2×RJ-45 10/100 Ethernet; protocols Modbus/TCP, DNP3 L2, IEC 104, MQTT; power 10–30 V DC (24 V @ 5 W typical); operating temperature –40 to 70 °C; DIN-rail or wall mount. ADAM-4024 (analog output, used for dosing pump control), ADAM-4050/4017+ and the ADAM-5000-series backplane (analog I/O cards ADAM-5017P, ADAM-5024) are confirmed in real pilot I/O wiring documentation alongside the ADAM-3600. A more recent (2023) confirmed remote-I/O bill of materials uses **Weidmüller UR20** modular I/O instead: a UR20-FBC-MOD-TCP-V2 Modbus/TCP fieldbus coupler, one UR20-16DI-P (16-channel digital input), one UR20-16DO-P (16-channel digital output), and six UR20-4AI-UI-16 (4-channel universal analog input) modules on a DIN rail — i.e. a Modbus/TCP-based I/O station with analog input as the dominant channel count, consistent with the many 4–20 mA sensors described in chapter 13.
- Control-box ordering options: **CB0** (PLC only), **CB1** (PLC + touchscreen HMI).
- Cloud-connectivity tiers: **CL0** (local only), **CL1** (cloud monitoring), **CL2** (full cloud monitoring and control) — see chapter 19.
- Incoming supply (pilot): 3×400 V AC (3P+N+PE), max. 25 A; 24 V DC control bus.

14.3 Operation

The PLC drives the automatic sequencing described throughout Part II (pump speed, valve switching, dosing) and surfaces process values and alarms to the HMI/dashboard (chapter 6).

14.4 Maintenance

See chapter 7 for the general schedule (dust/filter check on the IP55 enclosure, terminal torque check) and log template.

14.5 Troubleshooting

- **ADAM-3600 timeout / cannot reconnect after uploading a new PLC program:** browse to the device's IP address, log in to its web UI (default password 00000000), go to *Configuration* → *Reboot*. After reboot, retry the stop/connect sequence. Precondition: the device must respond to `ping` — if it does not, the fault is network/ power related, not a program-upload timeout.
- **Modbus-RTU communication issue:** use qModMaster with a CH340 USB-to-Modbus-RTU adapter to verify communication directly: confirm the assigned COM port, set RTU options to match the device (e.g. 9600 baud, 8 data bits, 1 stop bit, no parity), set the correct slave address, use FC1/FC2/FC3, set Base Address to 0, then read a known register to confirm communication. The built-in bus monitor can inspect Tx/Rx traffic for deeper diagnosis.
- **Analog signal fault (any 4–20 mA device wired through this cabinet):** see the bench-test procedure in section 13.4.